

↔ THE FUNDAMENTAL TUG-OF-WAR IN PHYSICAL AI

The Structural Conflict Between High-Capacity Semantic Deliberation and Relentless Physical Dynamics

🧠 THE AI & DELIBERATION REALM

Untrusted MPU / GPU Horizon (System 2 / System 1.5)

What It Demands:

- Maximum expressive capacity (7B–70B VLM parameters).
- Multi-view spatial attention & deep token contexts.
- Iterative diffusion denoising (10–100 unrolling steps).
- Bounded time to deliberate on open-vocabulary goals.

"Give me 500 ms to think for higher accuracy!"



⚙️ THE PHYSICAL & DYNAMICS REALM

Deterministic Real-Time Microcontroller (System 1)

What It Demands:

- Gravity pulls at 9.8 m/s^2 without waiting for inference.
- Kinetic momentum ($\mathbf{p} = m\mathbf{v}$) causes blind travel.
- Information freshness decays instantly ($\sigma_{\text{pos}} = \sigma_0 + v\Delta t$).
- Motor stator coils demand microsecond PWM switching.

"Act in 1 ms or the mechanism collides!"

🔧 HOW PHYSICAL AI RECONCILES THE CONFLICT: THE THREE ARCHITECTURAL CONTRACTS

1. **Proposal–Permission Privilege Split** (MPU proposes → MCU audits)
- 2. **Action Chunking** (Predicts H steps to amortize delay)
- 3. **Expiring Leases** (Auto-fallback on lag)